

Lemma. Let I be a proper ideal of an integral domain R .

If a non-constant monic polynomial $P(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ satisfies

$$\overline{P}(x) = x^n + (a_{n-1} + I)x^{n-1} + \dots + (a_1 + I)x + (a_0 + I) \in (R/I)[x]$$

cannot be factored into smaller degree polynomials ($\Leftrightarrow$ irreducible)

then $P(x)$ is irreducible in $R[x]$.

Proof. Suppose that $\overline{P}(x)$ cannot be factored, and $P(x)$ is reducible.

That is, $P(x) = a(x) \cdot b(x)$ where $a(x), b(x)$ are non-constant monic polynomials.

Then, $\overline{a}(x)$ and $\overline{b}(x)$ is the factor of $\overline{P}$, it is contradiction.

Eisenstein Criterion.

Let P be a prime ideal in the integral domain R .

Suppose that a non-constant monic polynomial

$$f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in R[x]$$

satisfies $a_{n-1}, \dots, a_1, a_0 \in P$ but $a_0 \notin P^2$, then f is irreducible.

Proof. Suppose that $f(x)$ is reducible, factorable as $f(x) = a(x)b(x)$,

where $a(x)$ and $b(x)$ are monic, degree smaller than $f(x)$. Write

$$a(x) = x^r + \alpha_{r-1}x^{r-1} + \dots + \alpha_1x + \alpha_0 \quad \text{and}$$

$$b(x) = x^s + \beta_{s-1}x^{s-1} + \dots + \beta_1x + \beta_0. \quad \text{Then, by the assumption,}$$

$$\overline{f}(x) = x^n + (a_{n-1} + P)x^{n-1} + \dots + (a_0 + P) = x^n = \overline{a}(x)\overline{b}(x).$$

Since P is prime, R/P is again the integral domain,

so either $\overline{\alpha}_0$ or $\overline{\beta}_0$ is zero. WLOG, let $\overline{\alpha}_0 = 0$.

Then, $\overline{\alpha}_1 \cdot \overline{\beta}_0$ must be zero, as the unique degree 1 coefficient of $\overline{f}$,

so $\overline{\alpha}_1$ or $\overline{\beta}_0$ is zero. If $\overline{\beta}_0$, then $\alpha_0, \beta_0 \in P \Rightarrow \alpha_0\beta_0 \in P^2$, contradiction

If $\overline{\alpha}_1 = 0$, then similarly, $\overline{\beta}_0 = 0$ or $\overline{\alpha}_2 = 0$. Inductively, if $\overline{\alpha}_1 = \dots = \overline{\alpha}_i = \overline{\alpha}_0 = 0$,

then $1 \cdot \overline{\beta}_0 = 0$, so, $\overline{\beta}_0 = 0$, or $\beta_0 \in P$. Now, $\alpha_0\beta_0 \in P^2$, thus proved.